treekv: a tree-structured KV cache in llama.cpp

Markdown + LaTeX · Llama-3.2-1B-Instruct Q4_K_M · llama.cpp 10a60cf, CPU only · Pinaksh, Reehan, Ishaan

Domain and method

- **Markdown:** 5 llama.cpp docs; units = sections by heading level.
- **LaTeX:** TeX Live sample2e + 4 math notes; units = sections, theorem/lemma/definition/proof/figure/table environments (pylatexenc). A proof becomes the child of the statement it proves.
- One sequence ID per leaf; every ancestor carries all its leaves' IDs, so llama.cpp's mask = the tree mask.
- Children start right after their parent: siblings share positions, every root-to-leaf path is gap-free.
- Subtrees under 32 tokens are merged (at most 254 leaves, keeping 2 sequences for the question and the layout check).
- Evict = `seq_rm(seq, p0, p1)` for the subtree's own sequence IDs.

Tree statistics

doc	tokens	depth	seqs	leaf min/med/max	markup
md/embedding	690	4	7	47/92/162	17%
md/llguidance	922	3	5	94/123/261	31%
md/mtmd	1545	3	3	143/190/688	38%
md/rpc	1378	3	6	48/287/416	9%
md/security	1579	4	8	43/152/275	20%
la/kv_cache	802	4	6	36/79/115	30%
la/online_softmax	716	4	6	19/75/112	26%
la/roofline	487	4	5	45/61/90	25%
la/sample2e	1732	2	2	563/916/916	32%
la/tree_mask	587	5	5	33/67/107	28%

Test results (3 runs each, all answers identical across runs)

Test 1 (edit B): 10/10 PASS Test 2 (evict C): 10/10 PASS

- Answers about A identical after every edit and every eviction, in all runs. First-token logits: edit **0.0e+00**; evict up to 4.4e-01 when C precedes A (llama.cpp reuses C's freed cells for the question, so the CPU arithmetic order changes), **0.0e+00** with the cache layout held fixed.
- KV memory freed: **115.22 MiB** measured vs 115.19 MiB expected (evicted tokens × 32 KiB; +12 bytes/cell bookkeeping).
- **Test 0:** two prompts in one batch = separate runs: **match** (12/26 tokens, logit diff 0.0e+00); changing sequence 1 moves sequence 0 by 0.0e+00.
- Finding: with 6-token prompts the runs differ by 0.66: CPU kernels tile 4 tokens at a time, so unaligned batches round differently; isolation stays exact.

CPU run times (median of 3) · Intel(R) Core(TM) i7-7700HQ CPU @ 2.80GHz, 4 threads

doc	tokens	prefill ms	edit B ms	full / edit	evict ms
md/embedding	690	10275	1033	9.9x	0.071
md/llguidance	922	17417	1997	8.7x	0.091
md/mtmd	1545	29990	3989	7.5x	0.178
md/rpc	1378	23976	3585	6.7x	0.083
md/security	1579	29326	2266	12.9x	0.287
la/kv_cache	802	14196	1966	7.2x	0.035
la/online_softmax	716	13720	2147	6.4x	0.078
la/roofline	487	8523	1321	6.5x	0.039
la/sample2e	1732	33062	17865	1.9x	0.223
la/tree_mask	587	10285	859	12.0x	0.077